Speech and Influence Dynamics in a Monetary Committee*

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*We thank the Agence de l'innovation de défense and CIEDS pour funding.
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1. Introduction

Many real-world policy decisions are made by committees rather than by individuals. By bringing together diverse viewpoints, committee decisions are often believed to be more accurate than those made by individuals.[1] Others, however, highlight reasons—such as conformity—that may sometimes lead groups to make erroneous judgments.

To understand how committee members interact and influence one another, this paper examines deliberations at the Federal Open Market Committee (FOMC) of the U.S. Federal Reserve, which is responsible for setting U.S. monetary policy. A unique feature of the FOMC is that meeting transcripts are released after five years. Another ideal feature is the FOMC's consistent agendas, enabling comparisons over time. As a result, we can employ textual analysis to examine dynamic (across-meeting) influence—that is, how new ideas (proxied by speech patterns) emerge, who introduces them, and how they are later adopted or dismissed by the group.

The FOMC is composed of seven politically appointed members—including the Fed Chair—and twelve representatives from the regional Federal Reserve Banks. Members also differ in various demographics, such as gender, education, and prior work experience. Moreover, some members have extensive experience with the committee, while others are recently appointed. Our first goal is to investigate the influence of individual members by identifying the characteristics of those whose contributions are both novel (i.e., whose speech patterns differ from those in previous meetings) and impactful (i.e., whose speech patterns are adopted in subsequent meetings), thus connecting to a broad literature in economics (e.g., Jackson, 2008) and psychology (e.g., Turner, 2005) studying social influence. To capture both dimensions of influence—novelty and impact—we build two measures of text similarity across meetings.[2] The first similarity measure is the degree of 'backward similarity' (BS), which reflects the extent to which individual i's speeches at meeting t are similar to speeches in previous meetings (up to a certain lag) by all committee members. The idea is that speeches with low BS can be considered 'novel.' [3] Next, we build a measure of 'forward similarity,'

[1] For example, see Condorcet (1785) and Galton (1901).

[2] In doing so, we draw on Kelly et al. (2021) who study breakthroughs in technological innovation.

[3] We adopt the standard TF-IDF method combined with cosine similarity. As in Kelly et al. (2021), we allow the IDF weights to evolve over time as some terms become more or less common throughout the years. A key difference from Kelly et al. (2021), who study novelty in patents, is that our data has a panel structure and we want to measure the similarity of the same individual at different time points (i.e., different meetings). Hence, instead of computing the similarities between a speaker and each of the other speakers, we compute the similarities of speakers in relation to the aggregate of all the speeches for a given meeting.

(FS) which evaluates how speeches in future meetings by all committee members are similar to an individual member's speeches at time t. It is important to note that, by itself, a high FS is not sufficient to indicate influence—it could simply reflect a common speech pattern that the speaker is following. To account for this concern, we define influence as the ratio between forward and backward similarity for each FOMC member at meeting t. A higher ratio indicates greater influence: novel speeches that also influence future discourse can be considered pivotal points in speech dynamics.

We stress that our measure of influence is based on the ability to change speech patterns, not necessarily opinions, which are more difficult to measure. This also implies that influence, as we define it, is not a zero-sum game, where one member's influence diminishes that of others.[4] On the contrary, influential members can jointly enrich the discussion. This leads to the second research question of the paper: to assess the diversity of discourse during FOMC meetings. Speech diversity can be thought of as a proxy for cognitive diversity (see Lix et al, 2022), which is often believed to be beneficial by introducing a wide range of perspectives and ways of thinking (Page, 2008; Alesina and La Ferrara, 2005). Discourse diversity will be measured by calculating the average cosine similarity within each meeting of all individuals with respect to the entire committee. We will explore whether the diversity of discourse reflects demographic differences among members, disagreement over policy actions, or other factors such as the Chair's openness to debate.[5] We will also analyze discourse diversity in the different sections of the FOMC meeting. Our hypothesis is that cognitive diversity is especially valuable in the initial phase, where members share their views on the state of the economy. However, it may not necessarily be beneficial in the second phase, where actual policy decisions must be selected and dissonant speech patterns might make it harder to reach agreement on the chosen alternative.[6]

[4]This differs from measures of social influence in networks (Jackson, 2008), which quantify how each agent affects the limiting belief in group interaction, with influences summing to one.

[5]For example, Ben Bernanke, former Fed Chair, emphasized that "policy decisions should be based on open-minded discussion, not the views of only one individual. I tried to foster debate in our meetings and I encouraged creativity—blue-sky thinking." We can therefore test whether we observe greater cognitive diversity in speech patterns after he became Chair.

[6]Blinder (2007, p. 121) writes that there is a subtle balance to be struck: an ideal monetary policy committee should be "individualistic enough to reap the benefits of diversity, yet collegial and disciplined enough to project a clear and transparent message."

2. Literature Review

3. How the FOMC Operates

4. Representing Text as Data

Consider a set M of chronologically ordered meetings. Each meeting $m \in M$ includes speeches by several individuals, who may speak multiple times within the same meeting. To divide the text into documents, we must choose the level of aggregation. We consider two levels in our analysis. First, at the *speaker–meeting level*, where we aggregate all statements made by an individual during a given meeting. In this case, a document in our corpus corresponds to a speaker–meeting *intervention*, denoted by $i \in I$. Second, at the *meeting level*, where we aggregate all statements made by all individuals during a given meeting m. In this case, a document corresponds to the complete *meeting transcript*.

We define a surjective function s that assigns to each speaker-meeting intervention i its corresponding meeting in the set of meetings M:

$$s : I \to M$$

Then, $|s^{-1}(m)|$ is the number of interventions at meeting m, and the total number of interventions (the dimensionality of set I) is given by

$$|I| = \sum_{m \in M} \left| s^{-1}(m) \right| .$$

Documents are represented in vector form.[7] To reduce the size of the vocabulary, we have lemmatized all words, remove punctuation and stop-words, such as articles ("the," "a") and conjunctions ("and," "or"). We also exclude rare words (appearing less than 20 times) and excluding very common words.[8] After these operations, let $|W|$ denote the size of the vocabulary.

We now describe the term frequency matrix, where each value represents the frequency of a word within a document. Since we consider two levels of aggregation, we define two separate matrices accordingly.

[7] Following the bag-of-words approach, a document is treated as an unordered collection of words, with word order ignored and only word frequency retained.

[8] The list of frequent words that we exclude is: think, 's, will, say, go, one, now, well, see, get, may, just, thank, can, also, make, look, seem, come, give, move, good, way, know, might, even, take, suggest, continue, still, really, first, use, like, person, talk, thing, show, two, want, need, put, yes, part, believe, least, rather, run, comment, many, something, sense.

First, suppose a document is a speaker-meeting intervention i. We construct a matrix $\mathrm{TF} \in \mathbb{R}_+^{|I|\times|W|}$ (Term Frequency), which captures how often each word appears in a given intervention. Each element $\mathrm{TF}_{i,w}$ of the matrix represents the frequency of word w in document i, and is defined as:

$$\mathrm{TF}_{iw} \equiv \frac{c_{iw}}{\sum_{k\in W} c_{ik}} \tag{1}$$

where c_{iw} is the raw count of word $w \in W$ in document $i \in I$. This measure adjusts for document length by normalizing the word count, and thus describes the relative importance of term w within document i.

Second, when we aggregate interventions by meeting—so that a document corresponds to the complete *meeting transcript*—we define an *aggregate* term frequency matrix $\mathrm{TF}^{\mathrm{agg}} \in \mathbb{R}_+^{|M|\times|W|}$. For a word w in meeting m, term frequency is

$$\mathrm{TF}^{agg}_{mw} \equiv \frac{c_{mw}}{\sum_{k\in W} c_{mk}} \tag{2}$$

where c_{mw} is now the raw count of word w in meeting m.

As is common in textual analysis, word frequencies are subsequently weighted to emphasize discriminative terms across documents. Greater weight is assigned to words that appear in only a few documents, as these are more effective at distinguishing those documents from the rest of the collection. Conversely, words that occur frequently across the entire corpus (e.g., "think," "rate," etc.) contribute less to discrimination and are therefore given lower weight. In the literature (see Jurafski and Martin, 2022, Ch. 6), the standard approach is to use inverse document frequency (IDF), a vector of size $|W|$ that assigns each word a single weight, independent of time. A key limitation of this method is that it ignores the temporal dimension: words that are common today may not have been common in the past. As a result, if an FOMC member introduces a new term (or concept) that is later widely adopted, its IDF weight will be low—since it appears in many later documents—leading to an underestimation of that member's influence.

Following Kelly et al. (2021), we account for the temporal dimension by introducing *backward* IDF weights. Again, the definition of these weights depend on the level of aggregation. Let a speaker–meeting intervention i be the unit of analysis. In this case, the backward-IDF matrix is denoted by $\mathrm{BIDF} \in \mathbb{R}_+^{|I|\times|W|}$ and for word w in intervention i, the backward-IDF values are defined as follows:

$$\mathrm{BIDF}_{wi} = \log\left(\frac{1 + \#\text{ documents before } s(i)}{1 + \#\text{ documents before } s(i) \text{ containing term } w}\right). \tag{3}$$

The numerator is the total number of interventions in the corpus that occurred prior to the current meeting, while the denominator is the number of those prior interventions in which term w appears.[9] As desired, common words that have appeared in most or all documents will have low weight in (3). Note that the BIDF weight depends on the meeting date $s(i)$, and that all interventions within the same meeting share the same BIDF values. Expression (3) contrasts with the standard IDF weights used in the literature, where the numerator is the total number of interventions in the entire corpus, and the denominator is the number of interventions in which the term w occurs. Consequently, the standard IDF weight is a vector of size $|W|$, and does not depend on the meeting date.

If instead the unit of analysis is a meeting transcript, the backward-IDF matrix is denoted by $\text{BIDF}^{\text{agg}} \in \mathbb{R}_{+}^{|M|\times|W|}$. For a word w in meeting m, the backward-IDF value is defined as in (3), but with $s(i)$ replaced by m.

4.1 Measuring Similarity

Instead of computing pairwise similarities between a speaker and each of the other speakers, we find it more interpretable and computationally less heavy to compute the similarities of speakers in relation to the aggregate of all the speeches for a given meeting (not necessarily on the same dates).

Suppose we want to compare an individual intervention i occurring at date $s(i)$ with a meeting m, occurring at date m. For each term w in intervention i, we define our equivalent of the "term frequency- inverse document frequency" as follows:

$$\text{TFBIDF}_{(w,i;m)} = \text{TF}_{iw} \times \text{BIDF}_{w,\tau}, \quad \tau \equiv \min(s(i), m). \tag{4}$$

Similarly, for each for word w in meeting m

$$\text{TFBIDF}^{agg}_{(w,m;i)} = \text{TF}^{agg}_{mw} \times \text{BIDF}^{agg}_{w,\tau}, \quad \tau \equiv \min(s(i), m). \tag{5}$$

It is important to note that for both interventions, we apply the BIDF corresponding to the *earlier* of the two. As in Kelly et al. (2021), this normalization is intended to better capture the impact of one intervention on subsequent speeches.[10] Also note that this version

[9] A value of 1 has been added to the numerator compared to the original paper to handle the first document and to avoid zero-division or negative values when a word appears in all documents. *(Note: This correction is not yet reflected in the code.)*

[10] The idea is that if a member introduces a new concept that later becomes widely adopted and influential, we want that term to retain a high IDF weight. Since we primarily compute distances between interventions that are relatively close in time, using the minimum date has only a minor effect.

of TF-IDF is more complex than in the standard case. In the standard TF-IDF, we have a single TF-IDF value for a term w in document i. In our case, however, we compute a TF-IDF value for each pair of documents being compared, resulting in a distinct matrix for each comparison.

The similarity between a meeting m and an intervention i is then computed by calculating the cosine similarity between the vectors $(\text{TFBIDF}_{(w,m;i)})_{w\in W}$ and $(\text{TFBIDF}_{(w,i;m)})_{w\in W}$, that is,

$$\rho_{i,m} = \frac{\text{TFBIDF}_{i\tau} \cdot \text{TFBIDF}^{agg}_{m\tau}}{\|\text{TFBIDF}_{i\tau}\|\|\text{TFBIDF}^{agg}_{m\tau}\|} \qquad \tau \equiv \min(s(i), m) \tag{6}$$

Since counts are non-negative, the cosine similarity $\rho_{i,j}$ lies within the interval $[0, 1]$. Interventions that use the exact same words in the same proportions will have a similarity of 1. Conversely, if two interventions share no common terms, their vectors are orthogonal, resulting in a similarity of 0.

4.2 Measuring Influence

We propose a measure of *influence across meetings*. We focus on influence across meetings rather than within a single meeting for two reasons. First, participants generally prepare their remarks and decide on their votes in advance.[11] Second, measuring influence within a single meeting could be biased by the predefined speaking order, with those who speak first potentially appearing more influential.

To build a measure of influence, we first propose a measure of novelty: how much the

[11] This is particularly true following the 1993 transparency reform. Laurence Meyer, a former Fed governor, observed that FOMC discussions are more likely to influence decisions at subsequent meetings rather than those of the current meeting: "Was the FOMC meeting just a ritual dance? No. I came to see policy decisions as often evolving over at least a few meetings. The seeds were sown at one meeting and harvested at the next. So I always listened to the discussion carefully because it could change my view, even if it could not change my vote at that meeting. Similarly, although in my remarks to colleagues it might have seemed that I was addressing the policy concerns and decisions of the day, in reality, I was often positioning myself—and my peers—for the next meeting." Other quotes stressing the importance of across metings influence. In this paper we study speech patterns across meetings; Indeed tehre is evidence that "You would make a point, maybe forcefully if necessary, at one meeting in the hope that it would sink in and by the next meeting . . . or the meeting after that, you are starting to persuade people. (Blinder)" and (Lindsey) "I viewed the discussion at the meetings not necessarily as useful for influencing the vote of that day but as influencing the way the thinking was going to go, down the [road], in future meetings." I'd say you have to think of the FOMC as a continuum. It's a discussion which is going on and on and on, so everyone is coming in knowing the parameters within which there might be agreement and they know what they said in the past and everyone's history, so I would say basically they go in pretty much thinking what is going to happen is not predetermined but can be read from the tea-leaves of the previous meeting or meetings."

speech patterns of a given individual are different (novel) compared to previous speeches by all members in prior meetings (within a certain time window ω). Following Kelly et al. (2021), we refer to this novelty as *backward similarity*:

$$BS_i^\omega = \sum_{m \in \mathcal{B}_\omega^i} \rho_{i,m} \tag{7}$$

where $\mathcal{B}_\omega^i$ is the set of meetings in the time window prior to the date of intervention i, that is, $\mathcal{B}_\omega^i = \{s(i) - 1, \ldots, s(i) - \omega\}$. Interventions with low backward similarity deviate from the speech patterns in previous meetings and are thus considered novel.

Next, again following Kelly et al. (2021), we measure the impact of an intervention in future meetings (again within a certain time window ω) by its *forward similarity*, defined as:

$$FS_i^\omega = \sum_{m \in \mathcal{F}_\omega^i} \rho_{i,m} \tag{8}$$

where $\mathcal{F}_\omega^i$ is the set of meetings in the time window after the date of intervention i, that is, $\mathcal{F}_\omega^i = \{s(i) + 1, \ldots, s(i) + \omega\}$. Interventions with low forward similarity are not very influential. Notice, however, that by itself, a large forward similarity is not sufficient for influence—it could be that there is a common trend and that the person is just following it. To account for this possibility, as a measure of influence, we take the ratio:

$$q_i^\omega = \frac{FS_i^\omega}{BS_i^\omega} \tag{9}$$

so that influential interventions are those that are diverse from previous meetings and are influential for subsequent meetings.

4.3 Measuring Speech Diversity

We can easily compute the similarity within meetings (using 6) and define $\rho_{i,s(i)}$ as the similarity of member i's intervention in meeting $s(i)$. We then average these similarities [recall that when we aggregate at the meeting level, i is not excluded – all members are included– so we are taking a similarity between i speeches and a meeting that also includes i speeches We may want to change the code to exclude the person].

This will allow us to see how speech diversity varies over time, possibly as a consequence of leadership style.

5. Econometric Analysis and Results

5.1 Influence and Experience

For our three similarity measures, we can rank committee members and investigate how these measures are related to members' personal characteristics. Results show that among FOMC members, the Fed Chairs generally have the highest BS and FS similarity—precisely because they summarize and, at the same time, guide the discussion. Due to their role as consensus builders, however, Chairs tend to have lower pivotality. Among other results, we find that female FOMC members have higher BS and FS measures compared to their male counterparts, and interestingly, women tend to be more pivotal than their male peers. Among the most influential members are Mr. Kroszner and Mr. Warsh. The latter is the frontrunner candidate to replace Powell and left the Board of Governors due to his disagreement with quantitative easing (QE). Mr. Kroszner, appointed by G.W. Bush was not reappointed after his term ended, the Democrats preferred a more loyal Governor (Dan Tarullo) during the financial crisis. To think about: why are BS and FS are so close, and ranking is so similar?

We then examine the impact of FOMC tenure on the development and persistence of word-use patterns.

The basic influence equation with experience is:

$$y_{it} = \beta_0 + \beta_1 e_{it} + X_{it}\lambda + \gamma_i + \varepsilon_{it}$$

where: y_{it} is our measure of influence (either backward or formward) of individual i at time t, e_{it} is the experience within the committee (number of attended meetings) of individual i at time t, X_{it} is a set of control variables.γ_i is the individual fixed effect (captures unobserved individual characteristics), ε_{it} is the error term. Two issues that may complicate the basic setting. First, similarities between an individual and the meeting as a whole will depend on whether fellow members are similar to i in terms of characteristics (ex, if i has a phd, do most members also have a Phd?). Second, estimating the effect of experience is complicated by selection – FOMC members, especially Board members, resign early maybe because they do not have much influence.

We find that as seniority grows, members not only build on past discussions (i.e., BS increases, see Figure 5b) but also become more influential in shaping future deliberations (i.e., FS increases as well, see Figure 5b).[12] This suggests that over time, members become

[12]The pivotality index, however, is not significantly related to seniority.

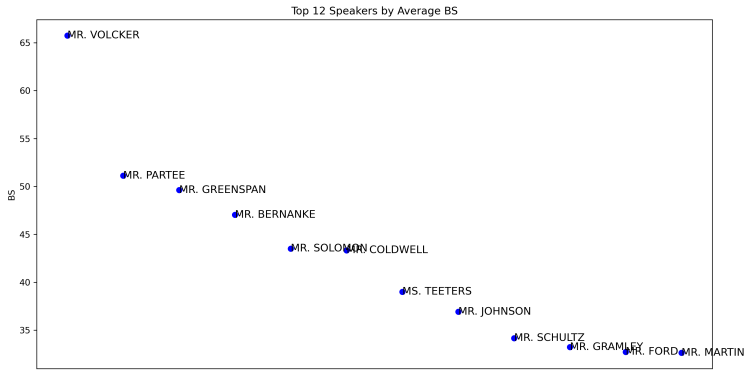


Figure 1: Top Backward similarity

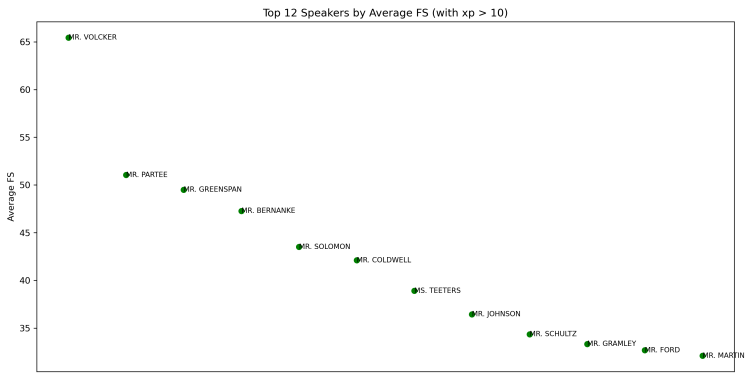


Figure 2: Top Forward similarity

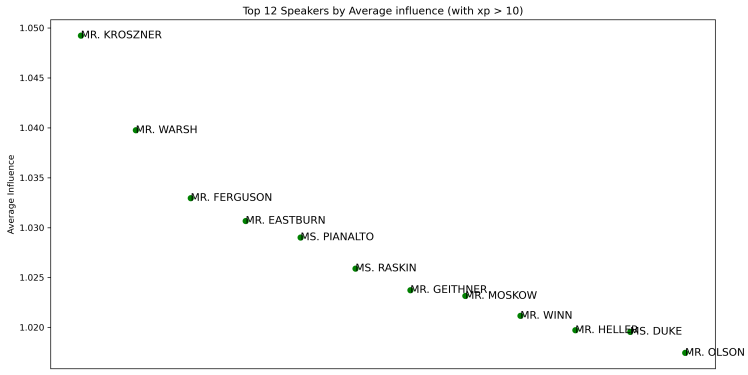


Figure 3: Top Influence

more integrated within the committee, potentially improving outcomes.

5.2 Speech differences among rookies

Let us consider the backward similarities of rookies. At their first meeting, by construction, members could not have participated in any previous meetings. Therefore, any positive backward similarity would not stem from direct interaction but rather from shared professional or institutional culture with other members.

We find that rookies differentiated by gender or by group (Bank President vs. Board) do not exhibit significantly different language. This suggests that these two characteristics are not discriminating factors in terms of linguistic patterns. It also suggests that linguistic diversity is not necessarily related to preferences for policy (hawks vs doves) given that Bank Presidents are known for being more hawkish than Governors. However, we do find evidence that members without an economics education tend to use different language than economists. Also, economists are more similar to the committee as a whole, reflecting the fact that trained economists represent the largest subgroup. Moreover, there are notable linguistic differences between members with academic backgrounds and those from business or banking, with academics displaying more similarity to past meetings.[13] This again reflects the fact that academics represent the largest subgroup.

The Figure 4 illustrate the different means for two characteristics (education and work experience soubgroups), and a simple Welch t-test indeed confirm that we can reject that they are equal.

Something to think about. Backward similarities may depend on the share of similar members – (since we are focusing on BS, we should look at (average) shares in previous meetings. Consider two groups, A and B (the shares of two group may add up to the total size of the group, but not always.) An individual belonging to group A has BS similarity

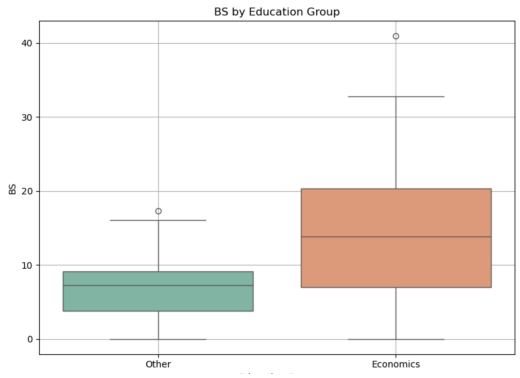
$$BS_{iA,t} = \gamma_{AA}\left(\frac{\#A}{\#tot}\right)_{\tau<t} + \gamma_{AB}\left(\frac{\#B}{\#tot}\right)_{\tau<t} + \phi_i + \varepsilon_{it} \tag{10}$$

which depends on the shares of the two groups in the committee. Similarly, for a member of group B

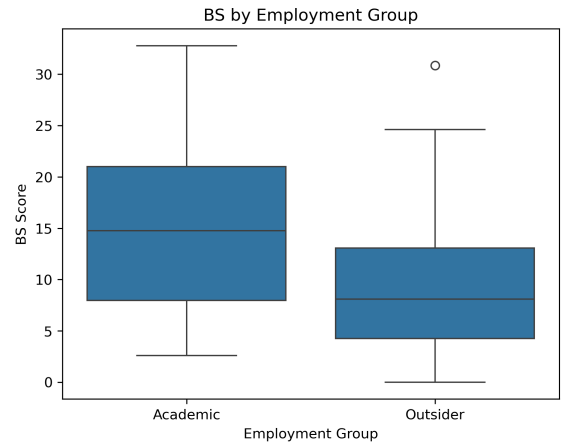
$$BS_{iB,t} = \gamma_{BA}\left(\frac{\#A}{\#tot}\right)_{\tau<t} + \gamma_{BB}\left(\frac{\#B}{\#tot}\right)_{\tau<t} + \phi_i + \varepsilon_{it} \tag{11}$$

A priori, we expect $\gamma_{AA}, \gamma_{BB} > 0$ and possibly –if the two groups have different speech patterns – $\gamma_{AB}, \gamma_{BA} < 0$. The sizes of these parameters should depend on within group

[13] Note that for work experience there is third group– those with Fed experience.



(a) Education: economics vs. non-economics



(b) Prior work experience

Figure 4: How rookies differ.

speech patters. For example, if group A is very homogeneous and has a very similar jargon γ_{AA} should be very high. Is there any (theretical) reason to expect γ_{BA} and γ_{AB} to be roughly the same?

With two groups $\#A + \#B = \#tot$,

$$BS_{iA,t} = (\gamma_{AA} - \gamma_{AB}) \left(\frac{\#A}{\#tot} \right)_{\tau<t} + \gamma_{AB} + \phi_i + \varepsilon_{it} \tag{12}$$

$$BS_{iB,t} = (\gamma_{BA} - \gamma_{BB}) \left(\frac{\#A}{\#tot} \right)_{\tau<t} + \gamma_{BB} + \phi_i + \varepsilon_{it} \tag{13}$$

5.3 Bernanke's blue-sky thinking experiment

Given our panel structure, we we estimate the effect of Bernanke's on speech diversity. In his autobiography, Bernanke wrote that "policy decisions should be based on open-minded discussion, not the views of only one individual. I tried to foster debate in our meetings and I encouraged creativity—blue-sky thinking." We can therefore test whether we observe greater cognitive diversity in speech patterns after he became Chair.

In our case, there is only one treatment period for all FOMC memebrs

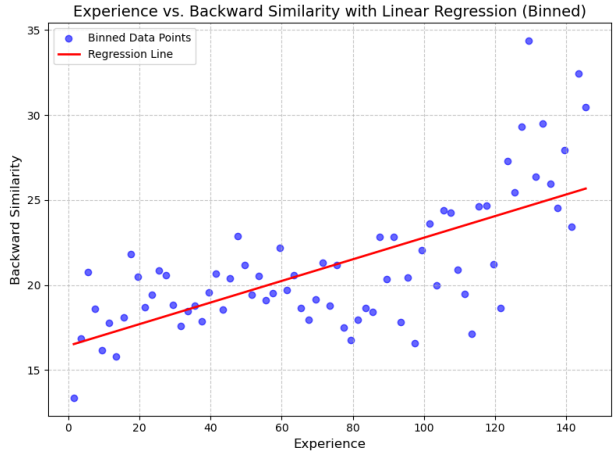
$$y_{it} = \sum_j \beta_j D_j + \gamma X_{it} + \lambda_t + \gamma_i + \varepsilon_{it}$$

where y_{it} is a measure of within meeting disocurse diversity, D_j is a dummy variable that takes value 1 if Benrnake was the chair and 0 if Greenspan was the chair at j meeting.

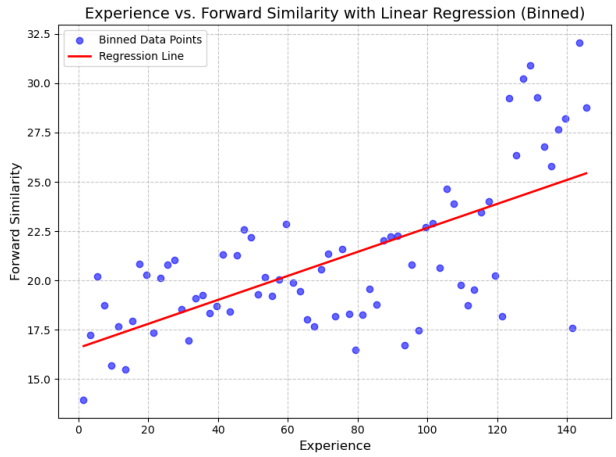
Time fixed effects is λ_t, capturing overall macro shocks. We also plan to use two different measures of diversity—one for the first part of the meeting (the economics round) and one for the policy round—to see whether open discussion in the first part (which seems beneficial to encourage more diverse viewpoints) leads to a more open, and possibly chaotic, discussion during the policy decision phase.

Indeed, lexical similarity has been shown to be associated with a broad range of benefits, such as better coordination on tasks (Darren et al, 2013 Nenkova et al, 2008), the self-reported quality of an interaction (Niederhoffer and Pennebaker, 2002), trust (Scissors et al, 2008), and group cohesiveness (Gonzales et al, 2010). However, while low text similarity may indicate a dysfunctional committee, it is unclear whether maximum lexical similarity is advantageous in a group of experts, as some diversity in speech patterns could help mitigate groupthink and herd behavior. As Blinder (2007, p. 121) argues, there is a subtle balance to be struck: an ideal monetary policy committee should be "individualistic enough to reap the benefits of diversity, yet collegial and disciplined enough to project a clear and transparent message."

Finally, Figures 6a and 6b illustrate that both measures of similarity fluctuate over time, displaying a downward trend. This could be an indication of increasing cultural differences and less effective deliberation, or, on a more positive note, as a sign of greater originality in discussions. To address this issue, the final part of the paper computes measures of deliberation quality (e.g., topic heterogeneity, obtained using LDA) and inflation preferences (relying on the RoBERTa-large model fine-tuned by Shah et al. (2023) to evaluate the hawkishness or dovishness towards inflation in FOMC statements) and relates these to our similarity measures.



(a) Backward Similarity



(b) Forward Similarity

Figure 5: FOMC Seniority and Similarity

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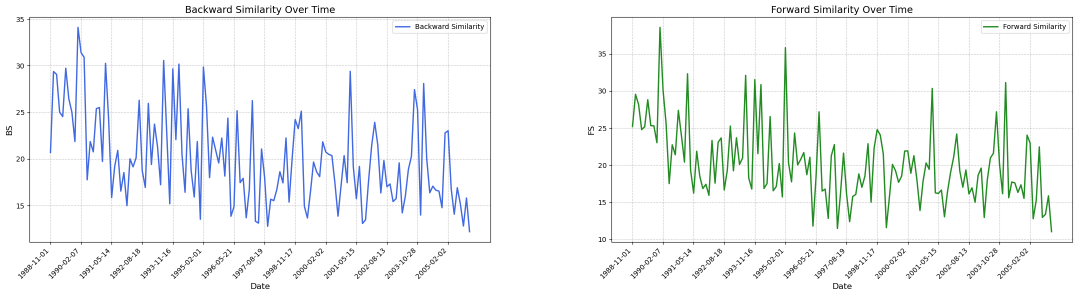

(a) Backward (b) Forward

Figure 6: Similarity over Time

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